

DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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Football Match Manager - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Football Match Manager is a structured data model designed to orchestrate the lifecycle of football matches, team management, and user tracking. It implements a relational pattern managing entities from Leagues down to User-Match follow relationships, ensuring strict data integrity through state-dependent validation rules for match statuses and scoring.

1.2 Maturity Assessment

The specification provides a high-fidelity data schema with exhaustive validation rules, yet it remains in REFINEMENT status. While the technical completeness score is high, critical structural gaps exist: the lack of defined high-level Goals & Objectives and a clear Scope boundary creates ambiguity regarding the system's business purpose and the exclusion of player/referee statistics.

1.3 Technical Stack

- **Identifiers:** UUID (128-bit unique identifiers)
- **Temporal Standard:** UTC (Coordinated Universal Time)
- **Data Types:** DateTime, Enum, String, Integer, Boolean

1.4 Architectural Constraints

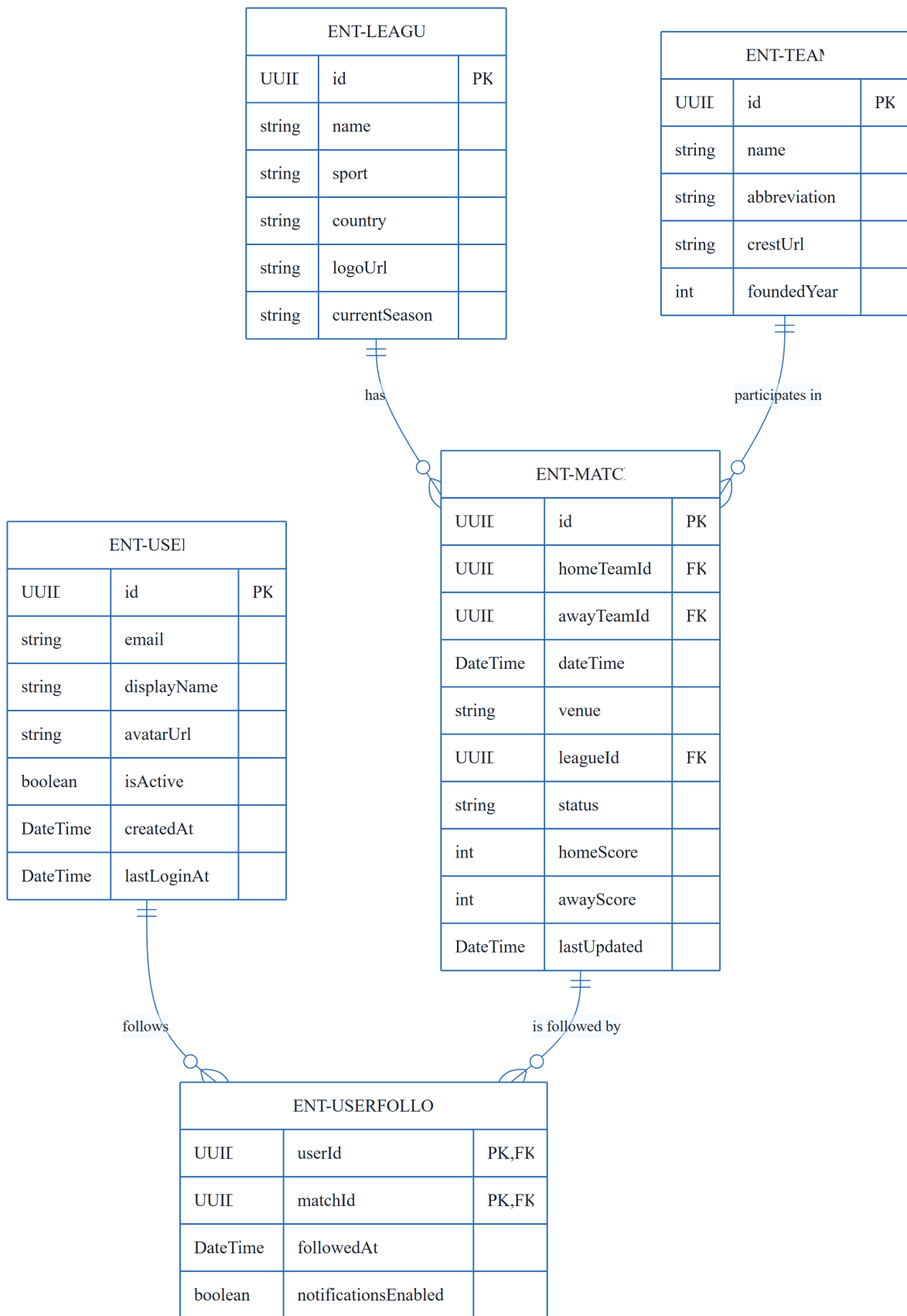
- **Entity Integrity:** Match `homeTeamId` must not equal `awayTeamId`.
- **Temporal Logic:** Match `dateTime` must be in the future when status is 'upcoming'.
- **State-Based Validation:** `homeScore` and `awayScore` must be non-null and ≥ 0 when status is 'finished'.
- **Uniqueness:** Team names must be globally unique; User emails must be unique.
- **Relational Constraints:** `UserFollow` must satisfy a composite unique constraint on (`userId`, `matchId`).
- **Domain Limits:** Team `foundedYear` must be between 1800 and the current year.
- **Field Constraints:**
 - Venue: max 200 chars.
 - Team name: max 100 chars.
 - Team abbreviation: max 5 chars.
 - Display name: max 50 chars.

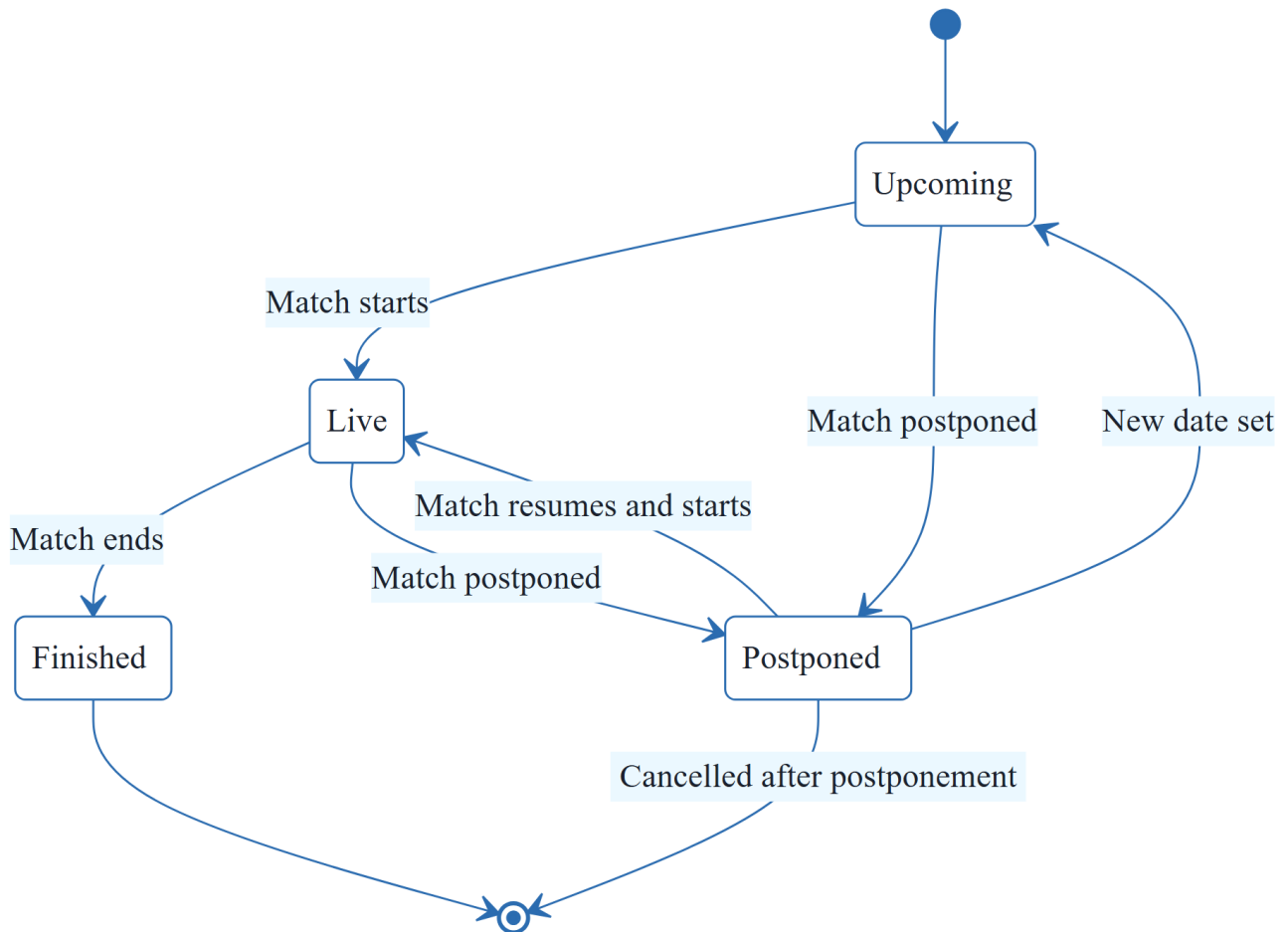
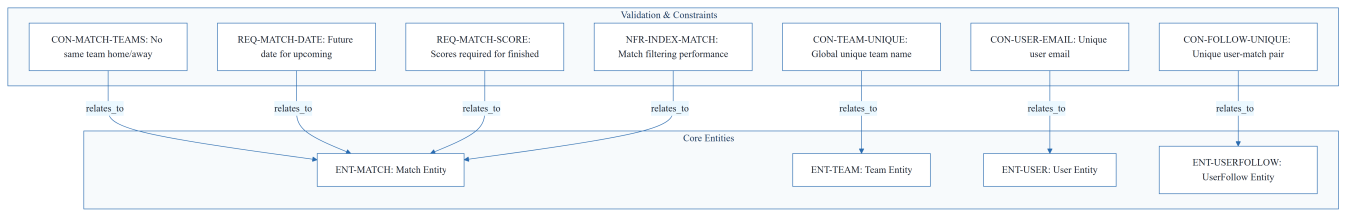
- **State Machine:** Match status is restricted to the Enum: upcoming, live, finished, postponed, cancelled.

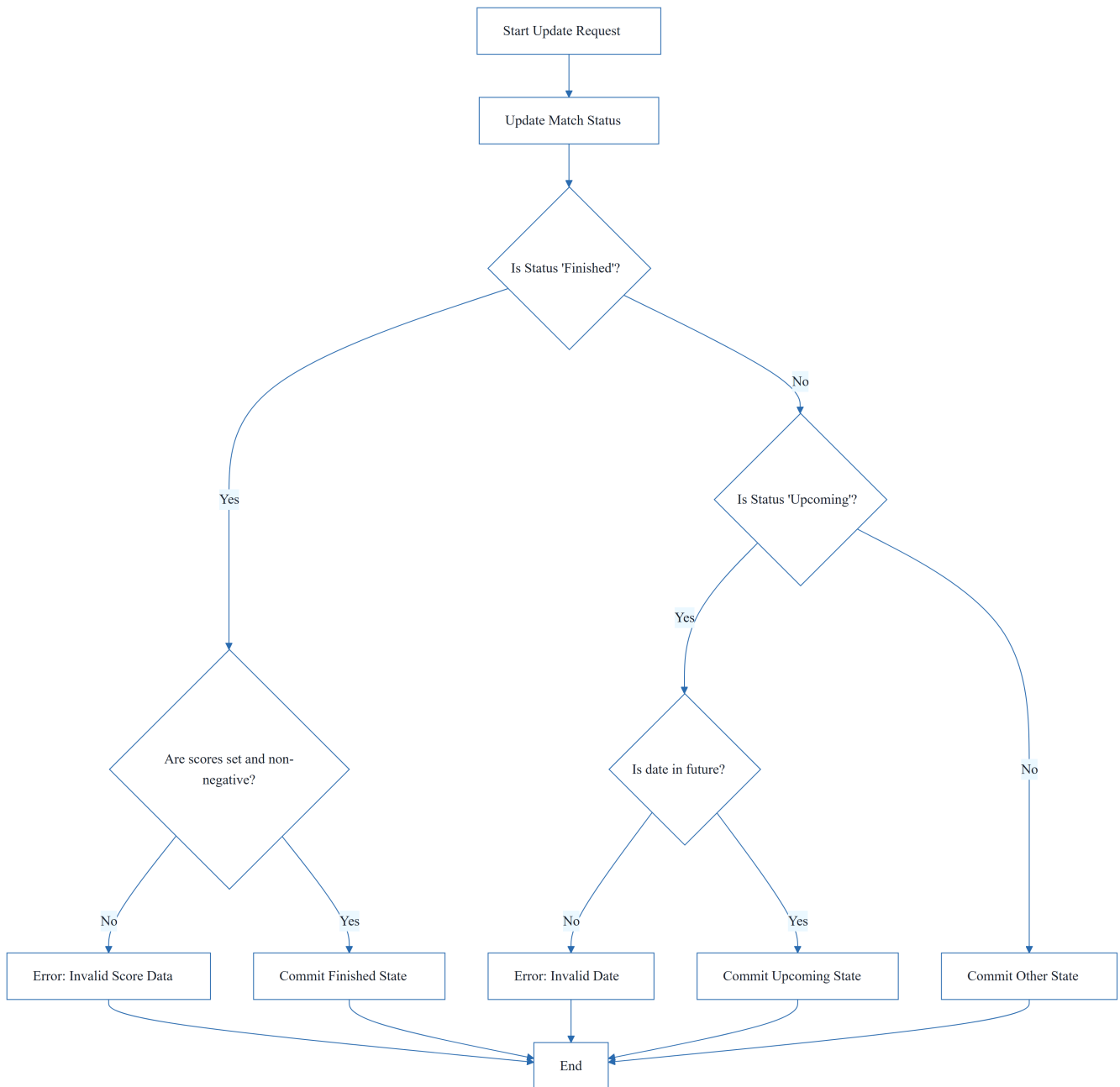
1.5 Critical Dependencies

- **Foreign Key Dependencies:**
 - ENT-MATCH depends on ENT-LEAGUE (leagueId) and ENT-TEAM (homeTeamId, awayTeamId).
 - ENT-USERFOLLOW depends on ENT-USER (userId) and ENT-MATCH (matchId).
- **Performance Index Requirements:**
 - ENT-MATCH: Index on (leagueId, dateTime) and (status).
 - ENT-TEAM: Index on (name).
 - ENT-USER: Index on (email).
 - ENT-USERFOLLOW: Index on (userId) and (matchId).

2. Architecture Workflows

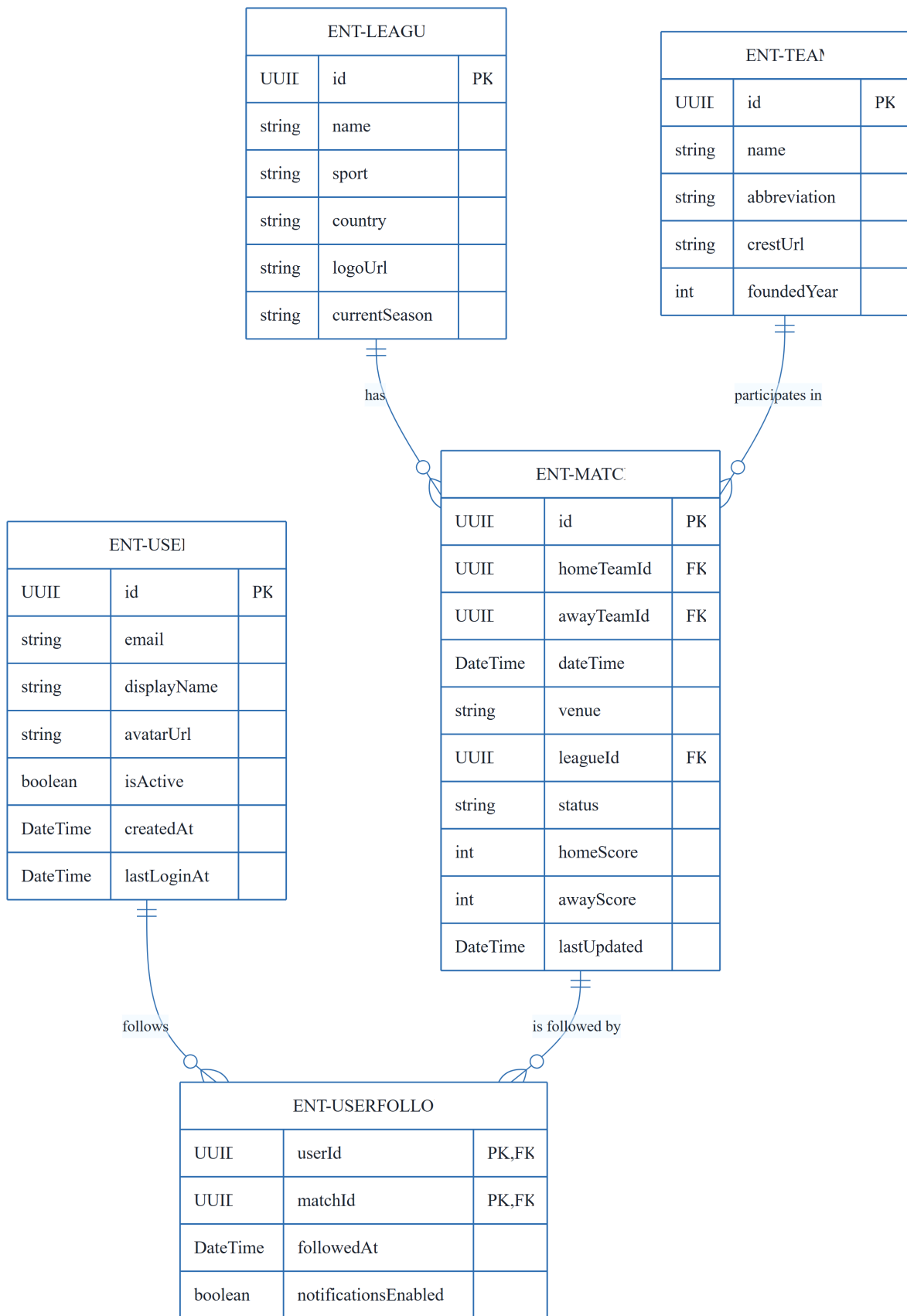




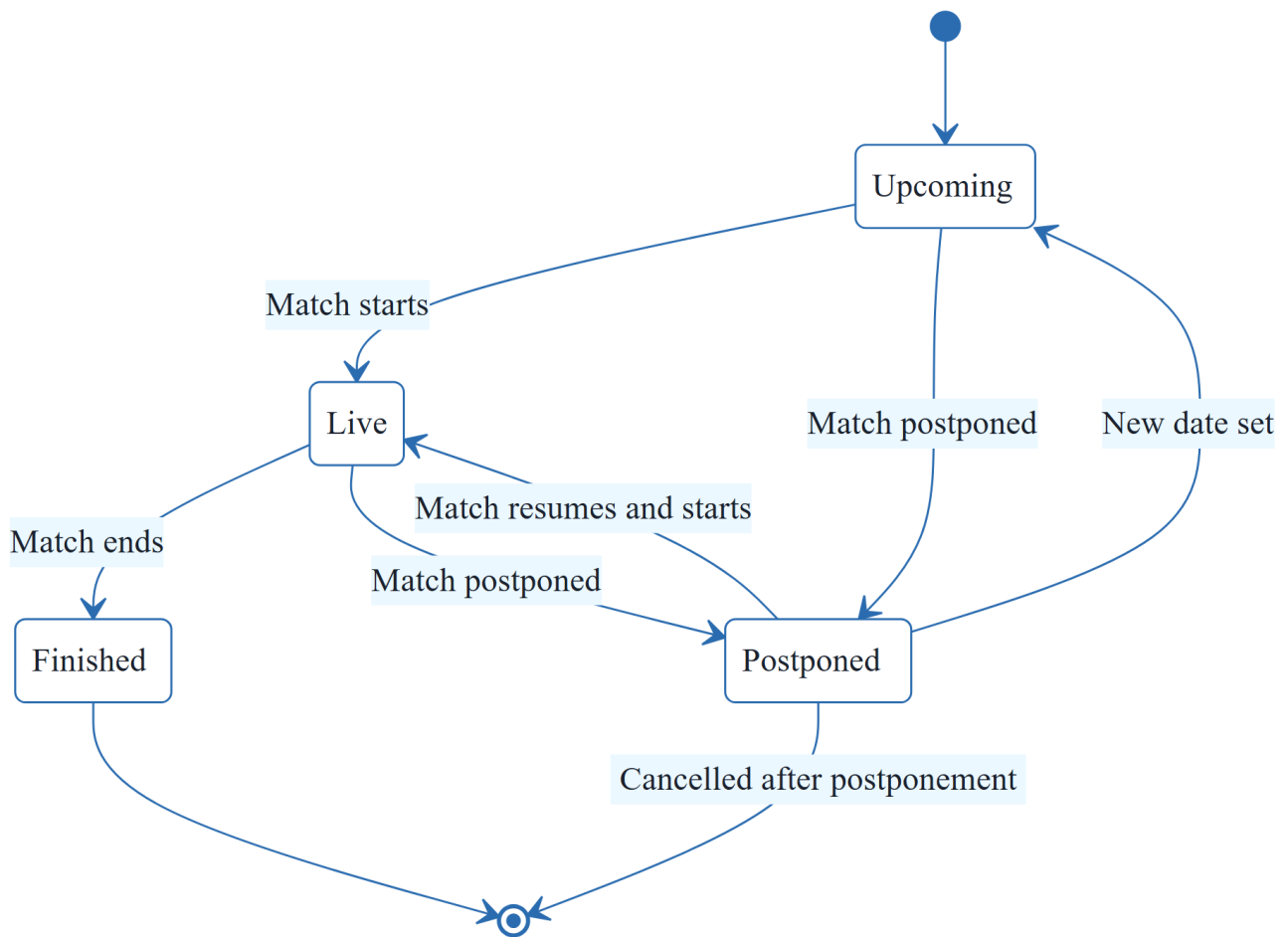


& Visual Diagrams

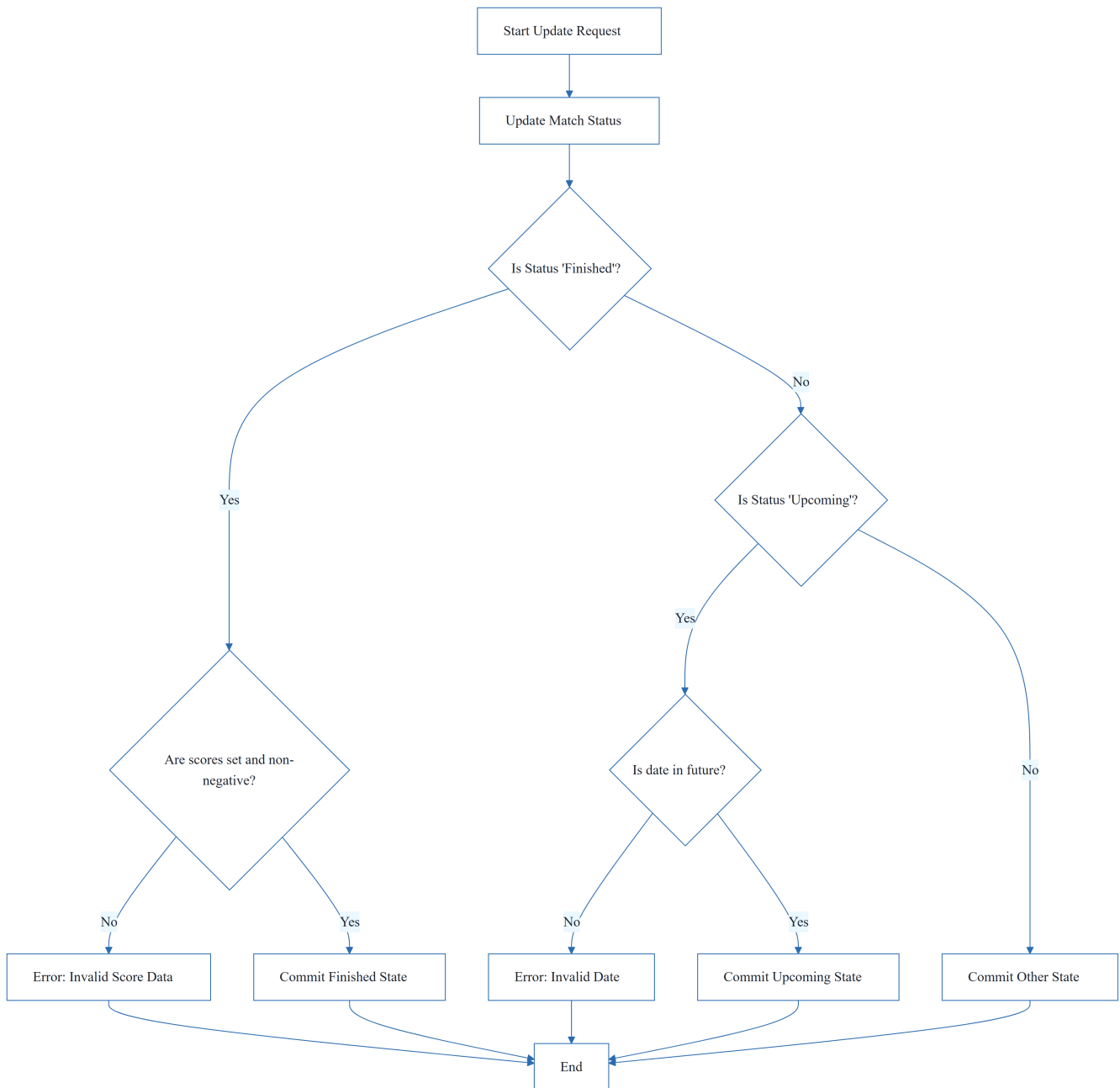
2.1 Entity Relationship Diagram



2.2 Match State Lifecycle



2.3 Match Update Workflow



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Type	Description	Target Entity
ENT-MATCH	Entity	Match entity: represents a football match with teams, league, date, and score.	N/A
ENT-TEAM	Entity	Team entity: represents a football team with name, abbreviation, and crest.	N/A

Identifier	Type	Description	Target Entity
ENT-LEAGUE	Entity	League entity: represents a football league or tournament.	N/A
ENT-USER	Entity	User entity: represents an application user.	N/A
ENT-USERFOLLOW	Entity	UserFollow entity: junction entity between User and Match.	N/A
CON-MATCH-TEAMS	Constraint	A match cannot have the same team as both home and away.	ENT-MATCH
REQ-MATCH-DATE	Functional	Match dateTime must be in the future for status 'upcoming'.	ENT-MATCH
REQ-MATCH-SCORE	Functional	When status changes to 'finished', both homeScore and awayScore must be set (non-null) and cannot be negative.	ENT-MATCH
CON-TEAM-UNIQUE	Constraint	Team name must be unique globally.	ENT-TEAM
CON-USER-EMAIL	Constraint	User email must be unique.	ENT-USER
CON-FOLLOW-UNIQUE	Constraint	Composite unique constraint on (userId, matchId) - a user can follow a match only once.	ENT-USERFOLLOW
NFR-INDEX-MATCH	Non-Functional	Index on (leagueId, dateTime) and (status) for Match filtering performance.	ENT-MATCH

3.2 Security Rules

- **Account Integrity:** User email must be unique (CON-USER-EMAIL).
- **Credential Security:** Password complexity requirements (min 8 chars) are mandated for authentication.
- **Data Access:** User account status is managed via the `isActive` boolean flag.

3.3 Data Models

ENT-MATCH

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
homeTeamId	UUID	Reference to Team	Required, exists
awayTeamId	UUID	Reference to Team	Required, exists, != homeTeamId
dateTime	DateTime	Match date and time (UTC)	Required, future if 'upcoming'
venue	String (200)	Stadium or location name	Optional
leagueId	UUID	Reference to League	Required, exists
status	Enum	Match status	Required, default: upcoming
homeScore	Integer	Goals scored by home team	Optional, min 0, default null
awayScore	Integer	Goals scored by away team	Optional, min 0, default null
lastUpdated	DateTime	Timestamp of last update	Auto-set on update

ENT-TEAM

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
name	String (100)	Team name	Required, globally unique
abbreviation	String (5)	Team abbreviation	Optional, unique per league
crestUrl	URL	URL to team crest/logo	Optional, valid URL
foundedYear	Integer	Year the club was founded	Optional, 1800 <= year <= current

ENT-LEAGUE

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
name	String (100)	League name	Required
sport	String	Sport type	Required, default: "football"
country	String (100)	Country where league is based	Optional
logoUrl	URL	URL to league logo	Optional, valid URL
currentSeason	String	Current season identifier	Optional

ENT-USER

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
email	String (email)	User email address	Required, unique, valid format
displayName	String (50)	Display name for the user	Required
avatarUrl	URL	URL to user avatar/image	Optional, valid URL
isActive	Boolean	Whether the user account is active	Required, default: true
createdAt	DateTime	Account creation timestamp	Auto-set on create
lastLoginAt	DateTime	Last login timestamp	Updated on login

ENT-USERFOLLOW

Field	Type	Description	Validation
userId	UUID	Reference to User	Required, exists
matchId	UUID	Reference to Match	Required, exists
followedAt	DateTime	Timestamp of follow	Required, auto-set on create
notificationsEnabled	Boolean	Notification preference	Optional, default: false

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Goals & Objectives	HIGH	Define the high-level purpose of the data model and what business goals it serves.
Scope & Out-of-Scope	MEDIUM	Clarify which football-related data is explicitly excluded (e.g., player stats, referee data).
Open Questions & Uncertainties	LOW	List unknowns, such as the exact password complexity requirements mentioned in the text.

4.2 Remediation & Workflow

The project is currently in the **REFINEMENT** phase. To move to a "Ready for Implementation" state, the project lead must address the high-priority gaps by defining the business objectives and establishing a clear boundary of scope to prevent feature creep (specifically regarding player/referee statistics).

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
Constraints	TECHNICAL_STACK	CON-FOLLOW-UNIQUE	Logical invariants enforced at the persistence layer to maintain data integrity, such as composite uniqueness requirements.
Cryptographic Hashing	TECHNICAL_STACK	CON-USER-EMAIL	The required transformation process for securing sensitive credentials before storage, implied by the complexity requirements for account access.
DateTime	TECHNICAL_STACK	ENT-MATCH	Temporal data type used to capture specific chronological points, including event schedules and system audit timestamps.
MUN	BUSINESS_DOMAIN	ENT-TEAM	An example of a short-form alphabetic representation of a professional club's identity.
Match Validation	BUSINESS_DOMAIN	Validation Rules	The set of operational checks ensuring a sporting event has distinct opposing sides and logically consistent timing and scoring states.
Relationships	TECHNICAL_STACK	ENT-MATCH	The structural associations and dependencies between different data entities, such as foreign key linkages.
Team Validation	BUSINESS_DOMAIN	CON-TEAM-UNIQUE	The verification process ensuring that professional club identifiers remain globally distinct within the registry.
UTC	TECHNICAL_STACK	ENT-MATCH	The standardized time offset used to ensure temporal consistency across different geographical zones.

Term	Category	Context Anchor	Project Definition
UUID	TECHNICAL_STACK	ENT-MATCH	A 128-bit numerically unique identifier used as a primary key for entity disambiguation.
User Validation	BUSINESS_DOMAIN	CON-USER-EMAIL	The integrity checks applied to account data, including the verification of unique electronic mail addresses and secret credential strength.
UserFollow	BUSINESS_DOMAIN	ENT-USERFOLLOW	The association entity representing a subscription link between an account holder and a specific sporting event.
UserFollow Validation	BUSINESS_DOMAIN	CON-FOLLOW-UNIQUE	The rule preventing duplicate subscription entries for the same account and event pair.